The idea of Democritus's system is extremely simple: the entire universe is made up of a boundless space in which innumerable atoms run. Space is without limits; has neither an above nor a below; is without a centre, or a boundary. Atoms have no qualities at all, apart from their shape. They have no weight, no colour, no taste. 'By convention is sweet, by convention bitter, by convention hot, by convention colour; but by verity atoms and void.'[5]

Atoms are indivisible; they are the elementary grains of reality, which cannot be further subdivided, and everything is made of them. They move freely in space, colliding one with another; they hook on to and push and pull each other. Similar atoms attract each other and join.

This is the weave of the world. This is reality. Everything else is nothing but a by-product, random and accidental, of this movement and this combining of atoms. The infinite variety of the substances of which the world is made derives solely from this combining of atoms.

When atoms aggregate, the only thing that matters, the only thing that exists at the elementary level, is their shape, their arrangement, and the order in which they combine. Just as by combining the letters of the alphabet in different ways we may obtain comedies or tragedies, ridiculous stories or epic poems, so elementary atoms combine to produce the world in its endless variety. The metaphor is Democritus's own.[6]

There is no finality, no purpose, in this endless dance of atoms. We, just like the rest of the natural world, are one of the many products of this infinite dance. The product, that is, of an accidental combination. Nature continues to experiment with forms and structures; and we, like the animals, are the products of a selection which is random and accidental, over the course of eons of time. Our life is a combination of atoms, our thoughts are made up of thin atoms, our dreams are the products of atoms; our hopes and our emotions are written in a language formed by combinations of atoms; the light which we see is comprised of atoms which bring us images. The seas are made of atoms, as are our cities, and the stars. It's an immense vision; boundless, incredibly simple and incredibly powerful, one on which the knowledge of a civilization would later be built.

On this foundation Democritus wrote dozens of books articulating a vast system, dealing with questions of physics, philosophy, ethics, politics and cosmology. He writes on the nature of language, on religion, on the origins of human societies, and on much else besides. (The opening of his *Little Cosmology* is impressive: 'In this work I treat of all things.') All of these books have been lost. We know of his thought only through the quotations and references made by other ancient authors, and by their summaries of his ideas.[7] The thought which thus emerges is a kind of intense humanism, rationalist and materialist.[8] Democritus combines a keen attention to nature,